

ORAL MINOXIDIL IN PATTERNED HAIR LOSS:
A CLINICAL PRODUCT MONOGRAPH

PRODUCT SUMMARY

Minoxidil or 2,4-pyrimidinediamine,6-(1-piperidinyl)-, 3-oxide¹: A hair growth enhancer²

- Minoxidil was introduced in the 1970s to manage severe refractory hypertension due to its vasodilatory effects.³
- Hypertrichosis emerged as a common side effect, prompting the development of a topical formulation, launched in 1986.³

Topical minoxidil has low patient compliance because of:

Unwanted
hair texture
changes³

Scalp rash
and pruritus⁴

Scalp scaling⁴

Formulation
odor³

Contact
dermatitis²

Allergy risks
(minoxidil or
excipients)³

Need for
twice-daily
application³



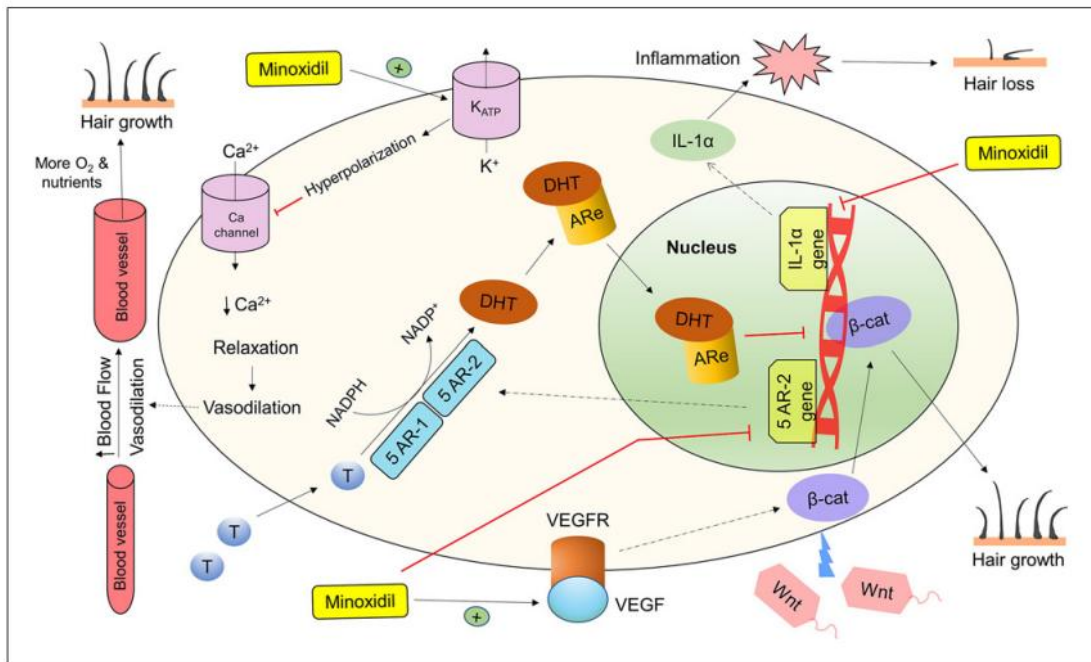
**Oral minoxidil
at low doses:**

Safe, effective, and well-tolerated
option for patients with patterned
hair loss with a good safety profile.³

- Oral minoxidil can circumvent the adverse events seen with topical therapy.²
- Once daily use may help improve its compliance compared to topical therapy.²

Mechanism of action of minoxidil

- Minoxidil is a prodrug. This is converted to its active form, minoxidil sulfate.²
- The positive effect of minoxidil on hair growth is mainly due to its metabolite, minoxidil sulfate, and the enzyme responsible for this conversion is sulfotransferase.²



Proposed mechanism of action of minoxidil⁵

5 AR-1, 5 alpha-reductase type I; 5 AR-2, 5 alphareductase type II; ARe, androgen receptor; β-cat, beta-catenin; Ca, calcium; DHT, dihydrotestosterone, IL-1α, interleukin-1 alpha; IL-1α gene, interleukin-1 alpha gene; K, potassium; NADP+, nicotinamide adenine dinucleotide phosphate; NADPH, reduced form of NADP+; O₂, oxygen; T, testosterone; VEGF, vascular endothelial growth factor; VEGFR, vascular endothelial growth factor receptor; Wnt, wingless-related integration site

Minoxidil as a Vasodilator⁵

- Minoxidil stimulates ATP-sensitive potassium channels, leading to vasodilation and increased oxygen and nutrients for hair growth.

Minoxidil as an Anti-Inflammatory Agent⁵

- May downregulate interleukin-1 alpha, reducing perifollicular microinflammation and facilitating hair growth.

Minoxidil as an Antiandrogen⁵

- May inhibit 5 alpha-reductase type II gene expression, acting as an antiandrogen.

Minoxidil as a Wnt/β-Catenin Signaling Pathway Inducer⁵

- May activate Wnt/β-catenin signaling via VEGF release in dermal papilla cells, promoting hair follicle regeneration.

OFF LABEL USE

Female-pattern
hair loss³

Male pattern
hair loss³

Chronic telogen
effluvium³

Traction
alopecia³

Loose anagen
syndrome³

Alopecia areata³

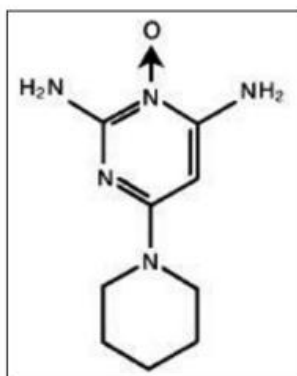
Monilethrix³

Chemotherapy-
induced hair
loss³

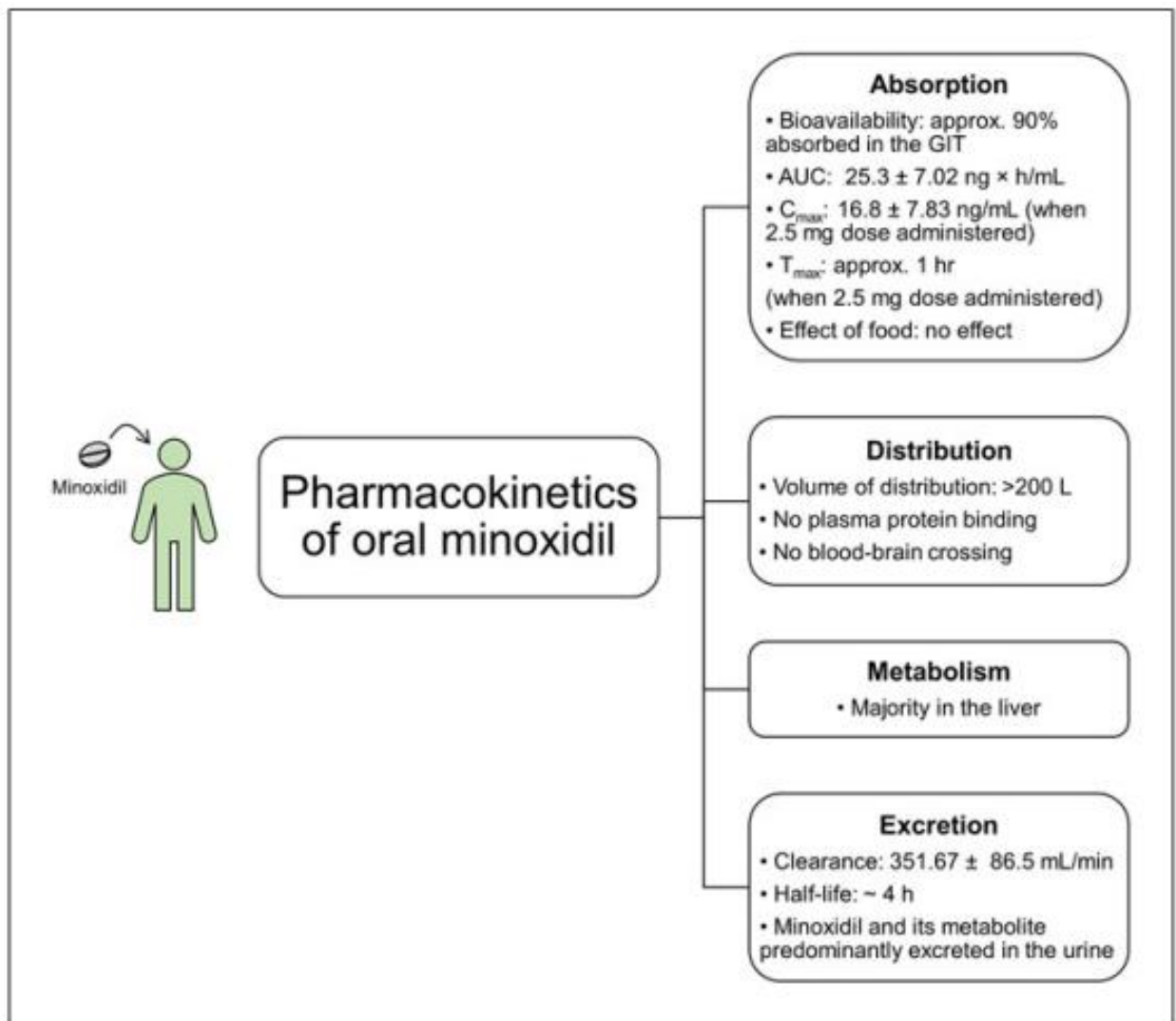
Scarring
alopecia³

- OM is an off-label therapy and its use for patterned hair loss should be individualized.³
- Patients who can benefit from OM are young adult patients with moderate patterned hair loss, or those with low compliance, local intolerance, or no response to topical minoxidil.³

PHARMACOLOGY OF MINOXIDIL



Structure of minoxidil²



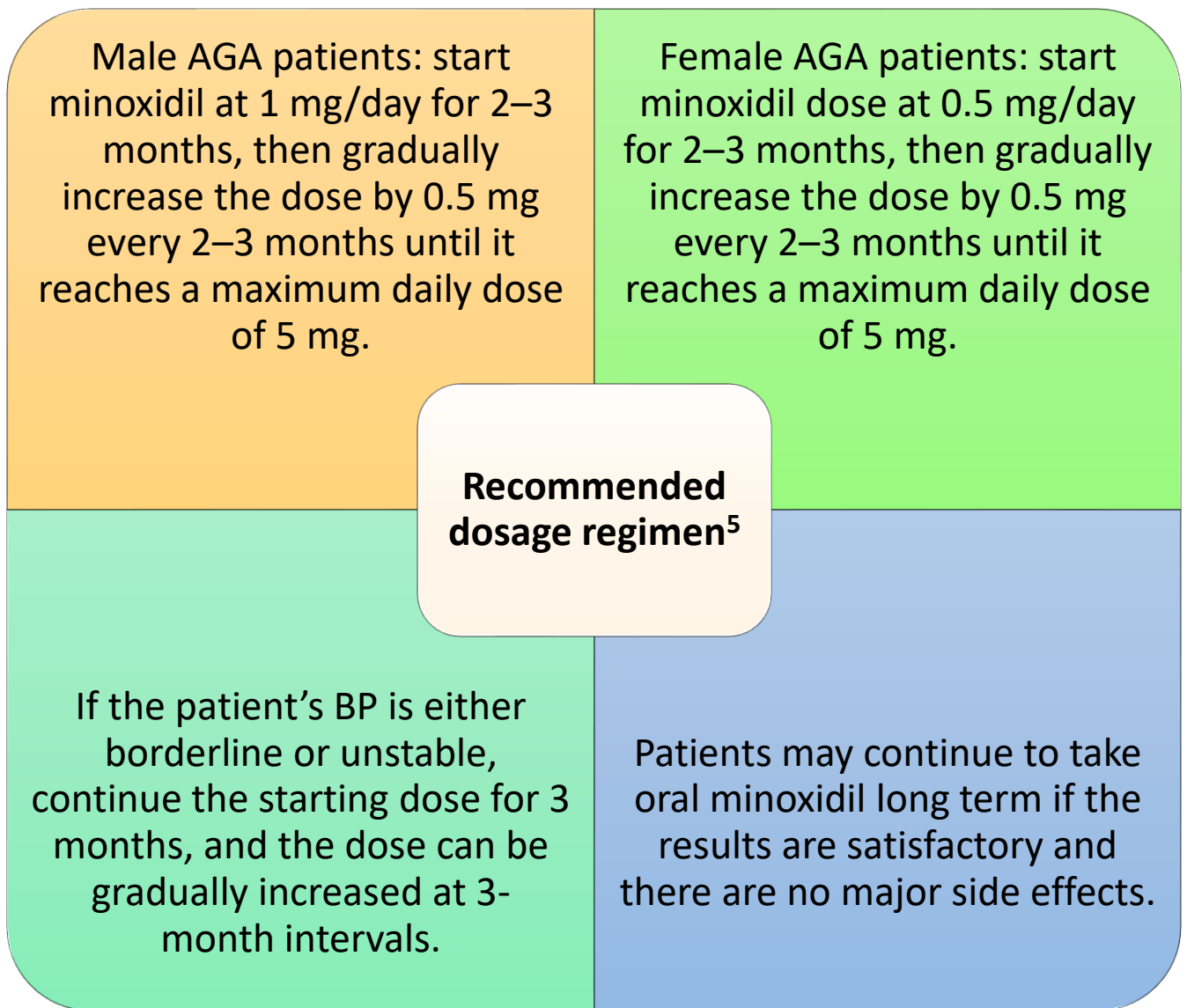
DOSAGE AND ADMINISTRATION

Female-patterned hair loss

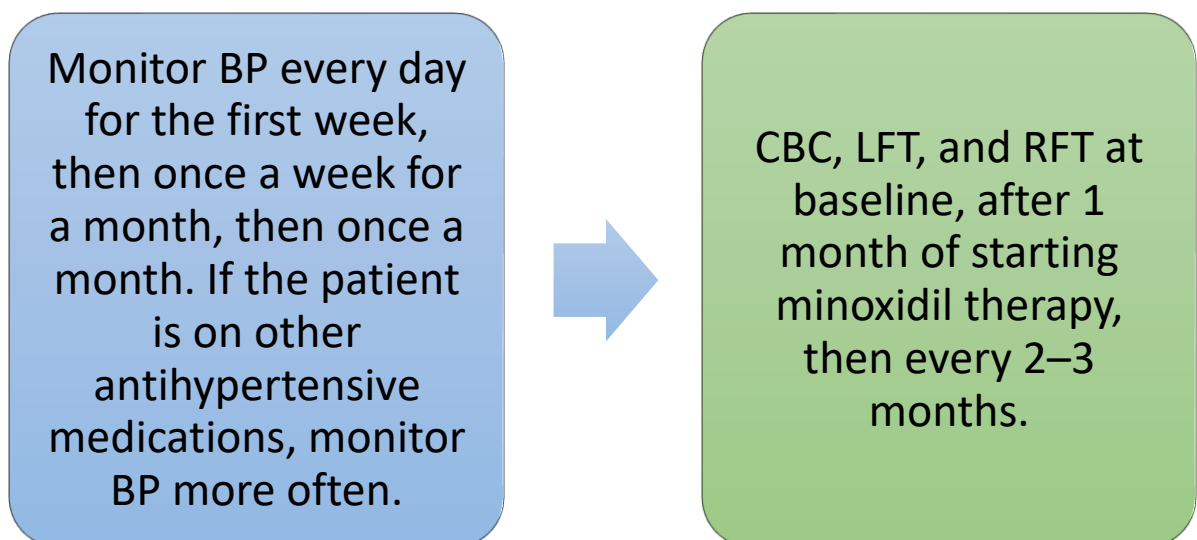
- OM, 0.25–1.25 mg daily, has been used for the treatment of FPHL, traction alopecia, and telogen effluvium, showing improvement in 61–86% of patients with a good safety profile.³

Male-patterned hair loss

- Effective treatment in males is generally seen with doses ranging from 2.5 to 5 mg/day. Overall, OM 5 mg/day is a very effective therapy for mild-moderate MPHL.³



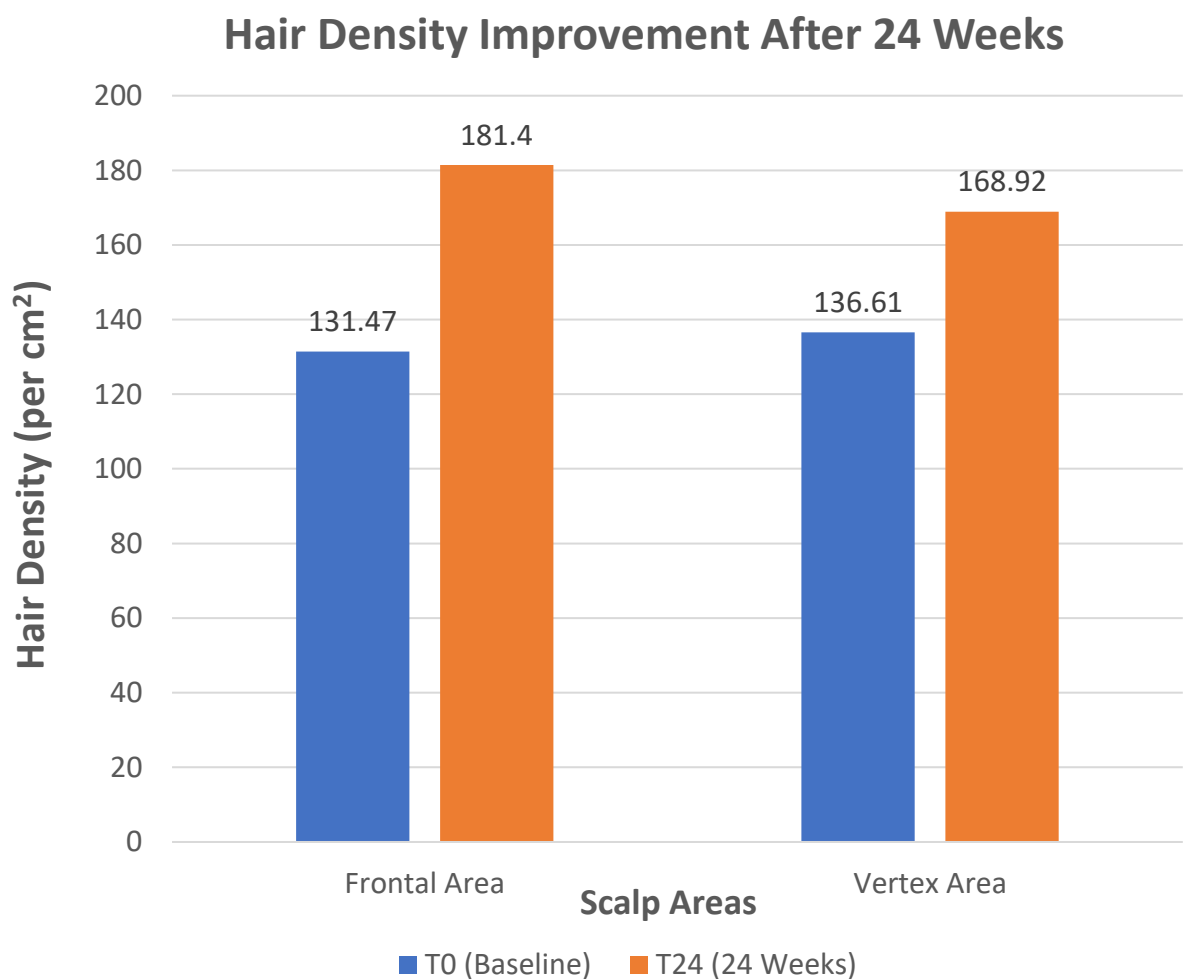
Monitoring guidelines⁵



EFFICACY PROFILE

A Study to Evaluate the Efficacy of Oral Minoxidil in the Treatment of Female Androgenetic Alopecia⁶

A 24-week retrospective study on 12 women (mean age 36.7 years) treated with 1.25 mg/day oral minoxidil

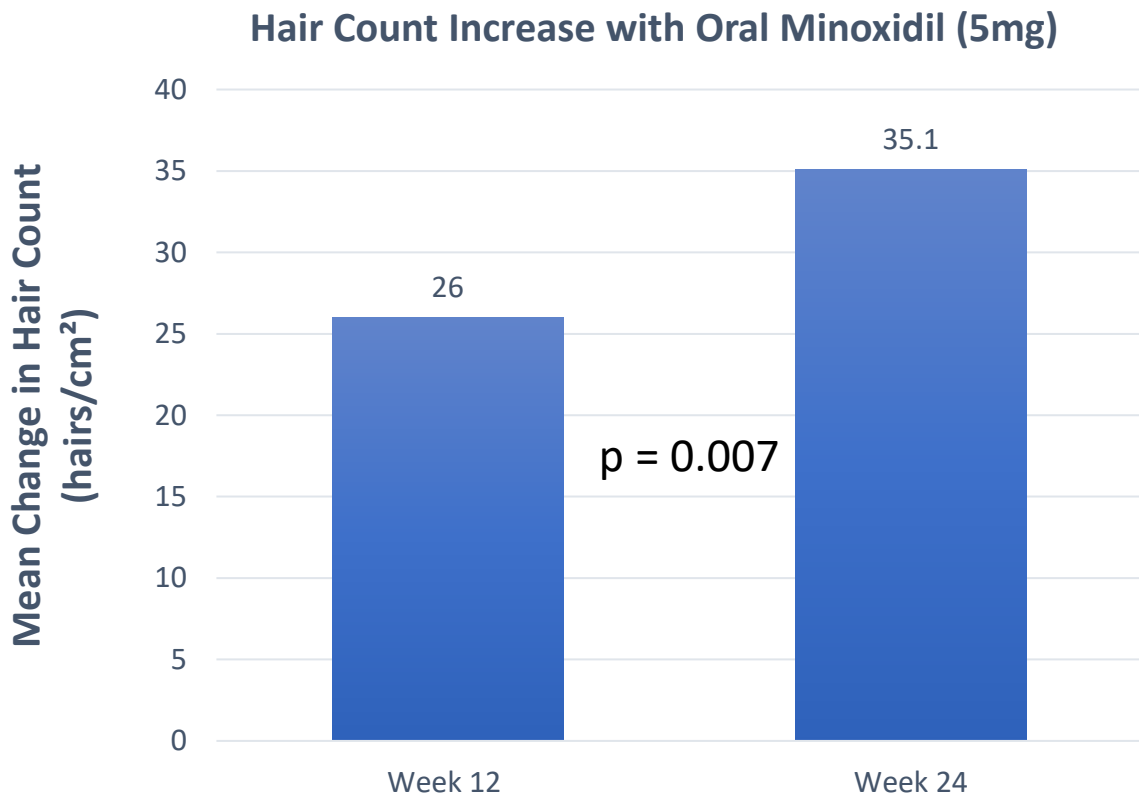


- After 24 weeks of treatment, a significant improvement in hair density was observed, with a 38% increase in the frontal area and a 23% increase in the vertex area.

In conclusion, oral minoxidil proves to be an effective and well-tolerated treatment for female androgenetic alopecia (AGA), even at higher doses (up to 2.5 mg/day).

A Study to Evaluate the Efficacy of Oral Minoxidil 5 mg in Male Androgenetic Alopecia (AGA)⁷

Study involving 30 men aged 24–59 years with AGA types III–V, treated with oral minoxidil 5 mg once daily for 24 weeks



Hair Count:

Significant increase in total hair counts at week 12 (mean change +26 hairs/cm²) and week 24 (mean change +35.1 hairs/cm²) (both $p = 0.007$).

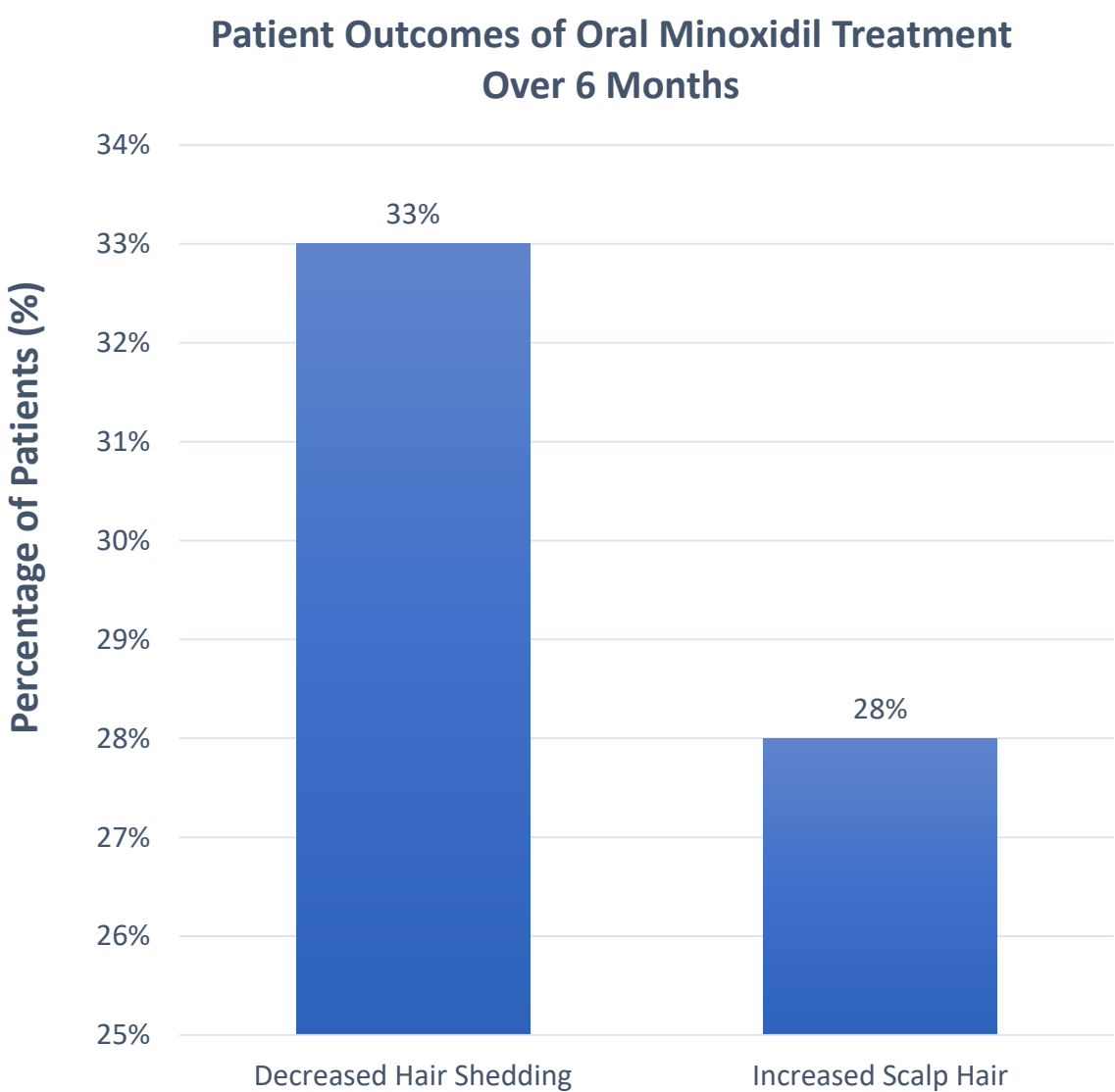
Photographic Assessment:

- 100% improvement in the vertex area (score > + 1).
- 43% of patients showed excellent improvement (score + 3, 71–100% increase).

Oral minoxidil 5 mg once daily was effective in promoting hair growth in male patients with AGA and had a favorable safety profile in healthy subjects.

A Study to Evaluate the Efficacy of Low-Dose Oral Minoxidil (1.25 mg) in Patients with Androgenetic and Traction Alopecia⁴

A 6-month evaluation of 1.25 mg daily oral minoxidil was conducted in 18 patients (17 females and 1 male) with AGA and traction alopecia



Low-dose oral minoxidil (1.25 mg daily) demonstrated efficacy in reducing hair shedding and increasing scalp hair in patients with androgenetic and traction alopecia.

SAFETY PROFILE

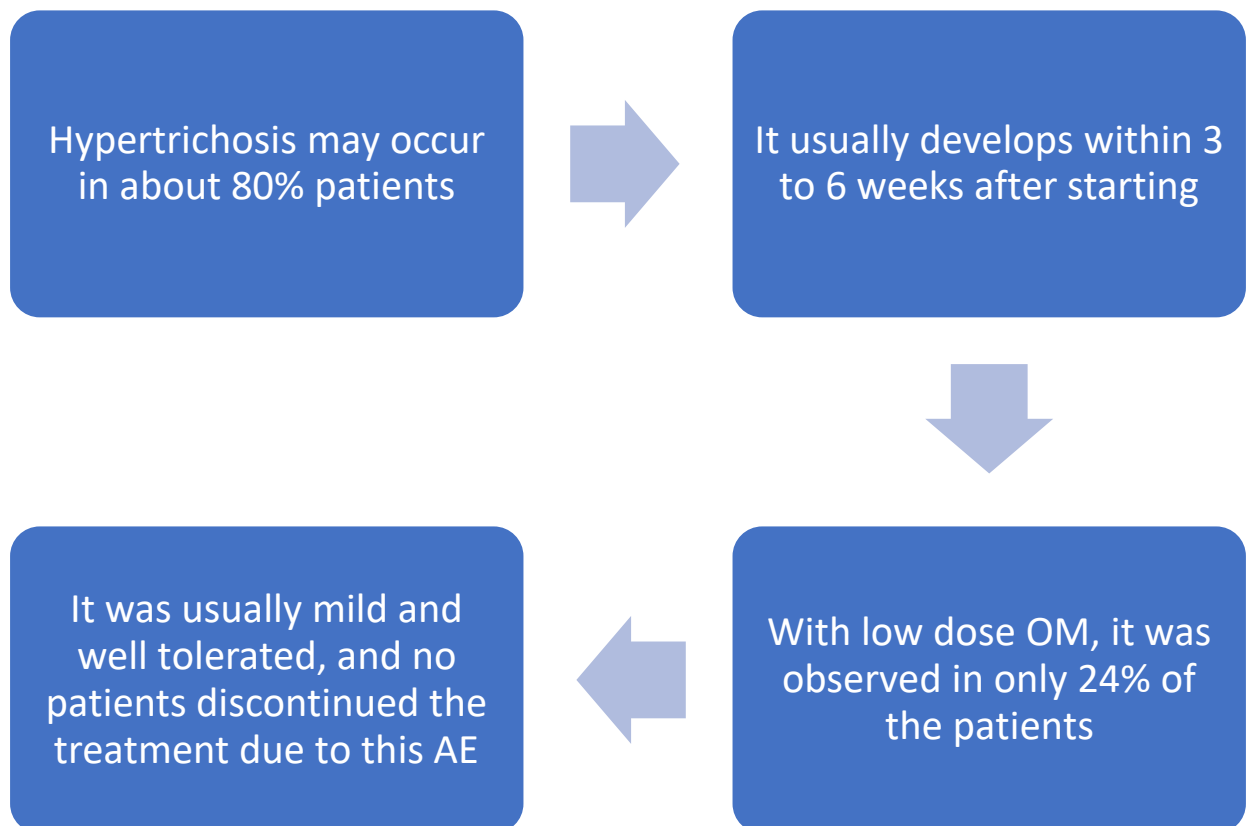
Successful treatment of various types of alopecia with low-dose oral minoxidil (OM), demonstrating a good safety profile has been reported⁸

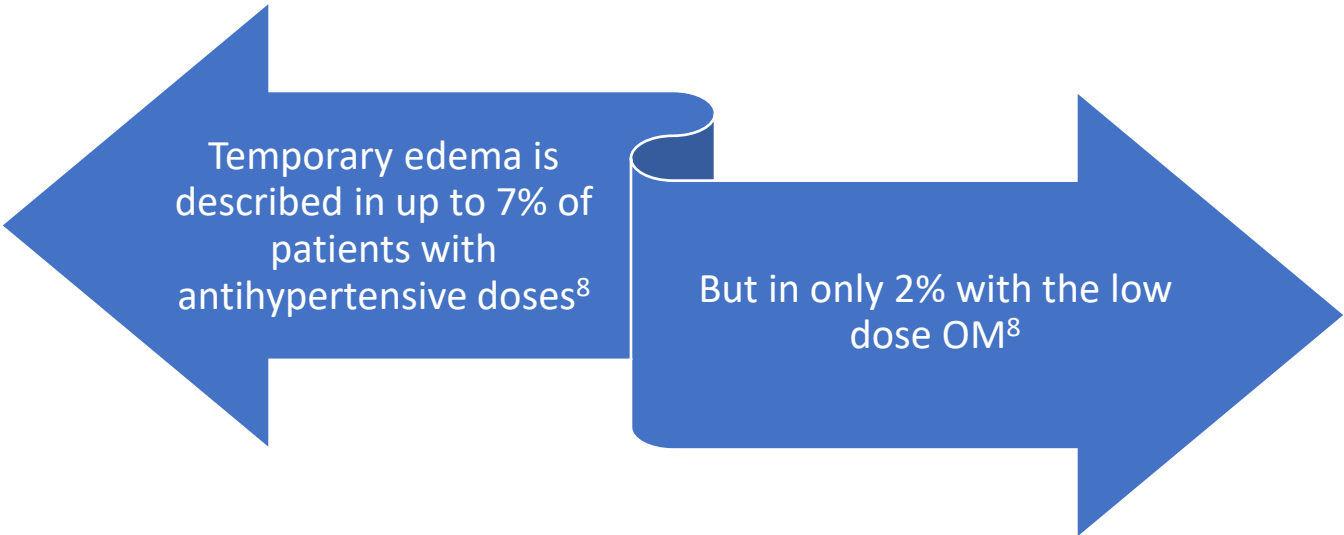


- OM is being increasingly used for the treatment of AGA and other types of hair loss⁸
- OM has a good clinical response rate, observed in 70% to 100% of the patients⁸

Adverse effects (AE) are the main concern that prevent the use of OM for the treatment of hair loss.⁸

According to the product monograph:⁸





Temporary edema is described in up to 7% of patients with antihypertensive doses⁸

But in only 2% with the low dose OM⁸

It usually occurs along with other predisposing factors such as prolonged standing or hot weather, but it may be responsible of discontinuation of the treatment.⁸

OM 2.5 and 5 mg in healthy volunteers have minimal effect on BP in normotensive individuals⁸

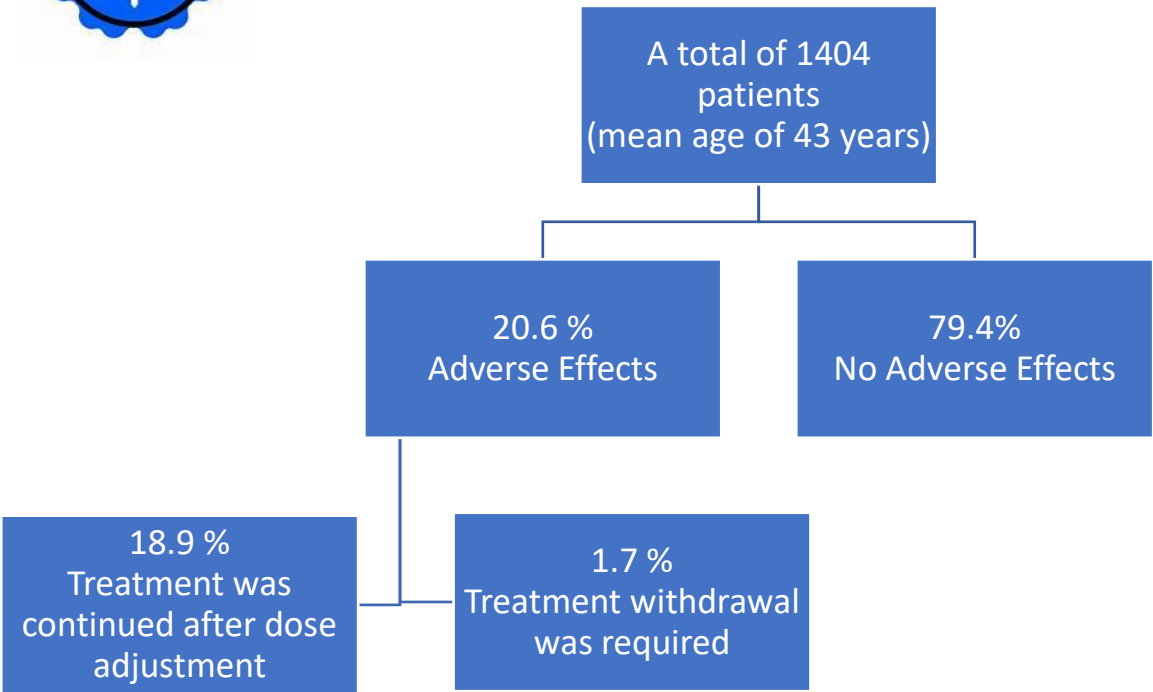
- ❖ Low dose OM has a good safety profile.⁸
- ❖ Hypertrichosis is the main AE, and it is usually mild and well tolerated.⁸
- ❖ More relevant AE such as pedal edema, postural hypotension and heart rate alterations are rare.⁸

Before OM, it's important to consider EKG changes and the risk of hypotension. A comprehensive patient history should include any past EKG alterations, blood pressure issues, or cardiovascular disorders, and a cardiologist's evaluation may be warranted in some cases.⁸

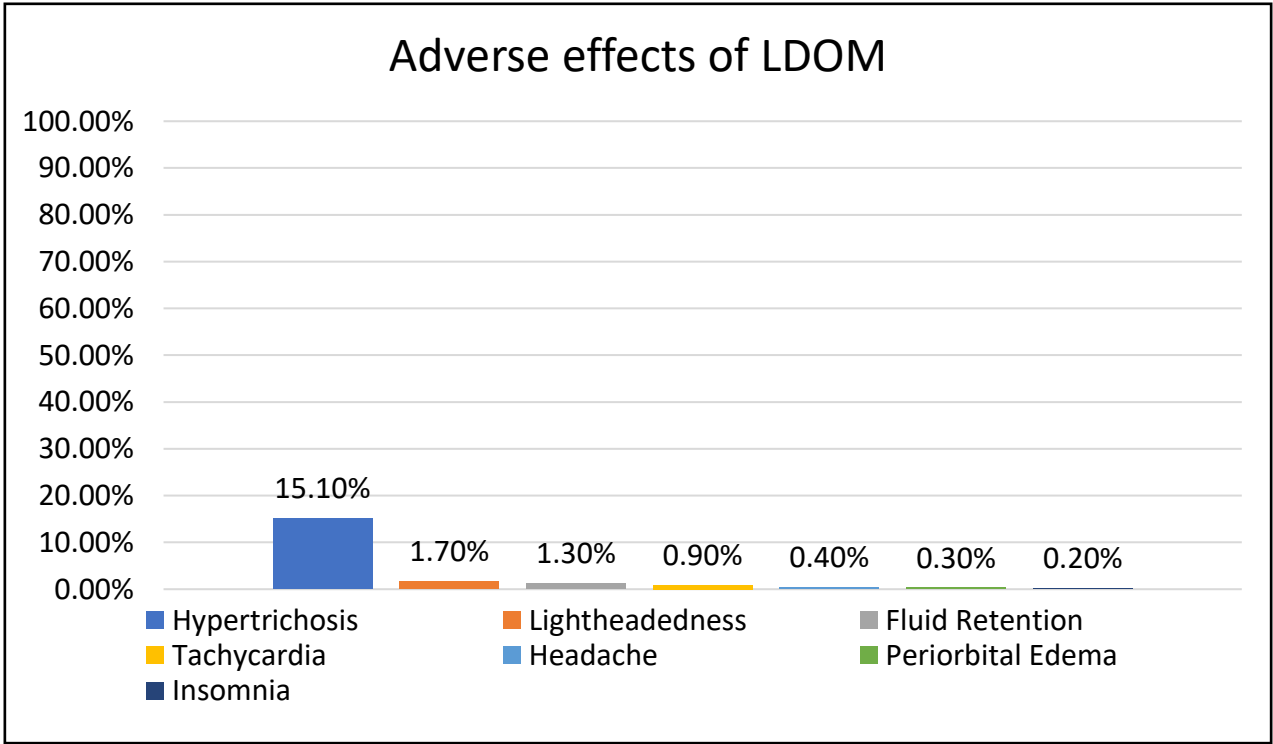
A study to describe the safety of LDOM for the treatment of hair loss in a large cohort of patients⁹



A multicenter study of 1404 patients



Adverse effects of LDOM



- No life-threatening adverse effects were observed
- The most frequent adverse effect was hypertrichosis (20%-24%), which rarely led to treatment discontinuation.

CONCLUSION

LDOM had a favorable safety profile in this large cohort of patients with various types of alopecia.⁹

Based on this study, the recommended doses and titration regimens for OM:⁹

- Females: Starting dose of 0.5 mg/day followed by 0.25 mg increments every 3 months depending on response and tolerability up to a maximum dose of 2.5 mg
- Males: Starting dose of 2.5 mg/day followed by 1.25 mg increments every 3 months depending on response and tolerability up to a maximum dose of 5 mg/day.
- This protocol is recommended for patients aged 12 years, although lower starting doses should be considered in those aged 12-17 years.
- In patients with chronic hair loss conditions, low-dose oral minoxidil (LDOM) may be used indefinitely; otherwise, any improvement in hair density may be lost within a few months after discontinuation of treatment.

OM, 0.25-1.25 mg daily, has been used for the treatment of female androgenetic alopecia, traction alopecia, and telogen effluvium, showing improvement in 61 to 86% of patients and a good safety profile.¹⁰

CONTRAINDICATIONS AND PRECAUTIONS



- ☐ Contraindicated in patients with a known history of hypersensitivity to minoxidil or its constituents, including propylene glycol.¹
- ☐ Contraindicated in patients under 18 years of age.¹
- ☐ Contraindicated in patients with pheochromocytoma.²
- ☐ Not recommended for use in cases of sudden, uncertain, patchy hair loss, hair loss after childbirth, scalp infections, or inflammation.¹
- ☐ The product intended for males is not recommended for use in females, particularly those with hirsutism or a history of laser epilation due to the risk of hypertrichosis.^{1,2,5}
- ☐ Not recommended for use during pregnancy (category C) or breastfeeding due to potential risks, including neonatal hypertrichosis.³
- ☐ Not recommended for elderly patients with a risk of myocardial infarction, heart failure, chronic renal failure, or severe hypertension.³
- ☐ Use cautiously in patients with fluid retention issues, cardiovascular conditions, or renal conditions.²
- ☐ Should be used carefully in patients receiving other antihypertensive medications.²

SUMMARY AND KEY TAKEAWAYS

KEY TAKEAWAYS

- ✓ Oral minoxidil (OM) is a well-tolerated, effective treatment for various types of alopecia, including female and male-pattern hair loss, showing high clinical response rates (70–100%).⁸
- ✓ OM offers an advantage over topical minoxidil with its once-daily dosing, improving patient adherence and minimizing issues such as irritation and odor.^{2,3}
- ✓ OM can be safely and effectively used for the treatment of female androgenetic alopecia, traction alopecia, and telogen effluvium. Studies have shown improvements in 61–86% of female patients using doses of 0.25–1.25 mg daily.¹⁰
- ✓ OM is particularly beneficial for patients who cannot tolerate topical minoxidil or have not had success with other treatments.³
- ✓ Although OM is not FDA-approved for hair loss, it is commonly used off-label in clinics for treating AGA. Its off-label use has gained clinical acceptance due to promising results in improving hair growth.⁵

GENERAL PATIENT WARNING/CONSUMER INFORMATION



- ❖ Low-dose oral minoxidil (LDOM) should only be used under the supervision of a healthcare professional.²
- ❖ Common side effects may include postural hypotension, headache, giddiness, hypertrichosis, and pedal edema. If these symptoms persist or worsen, contact healthcare provider immediately.²
- ❖ Patients should immediately contact their physician if they develop ankle edema (swelling) or any other cardiovascular adverse events while using low-dose oral minoxidil therapy.²
- ❖ LDOM is not FDA-approved for all types of hair loss, but it is commonly used off-label for androgenetic alopecia. Its use should be carefully considered based on individual health conditions.⁵

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